

Redhat system admin 2

lab 1

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task 1:

1. Use systemctl to view the status of all the system services

```
amiralsayed@amir:~$ systemctl list-units --type=service
```

UNIT	LOAD	ACTIVE	SUB	DESCRIPTION
accounts-daemon.service	loaded	active	running	Accounts Service
alsa-state.service	loaded	active	running	Manage Sound Car
atd.service	loaded	active	running	Deferred executi
auditd.service	loaded	active	running	Security Audit L
avahi-daemon.service	loaded	active	running	Avahi mDNS/DNS-S
bluetooth.service	loaded	active	running	Bluetooth service
colord.service	loaded	active	running	Manage, Install
crond.service	loaded	active	running	Command Scheduler
cups.service	loaded	active	running	CUPS Scheduler
dbus-broker.service	loaded	active	running	D-Bus System Mes
dracut-shutdown.service	loaded	active	exited	Restore /run/ini
firewalld.service	loaded	active	running	firewalld - dyna
fwupd.service	loaded	active	running	Firmware update
gdm.service	loaded	active	running	GNOME Display Ma
gssproxy.service	loaded	active	running	GSSAPI Proxy Dae
irqbalance.service	loaded	active	running	irqbalance daemon
kdump.service	loaded	active	exited	Crash recovery k
kmmod-static-nodes.service	loaded	active	exited	Create List of S
libstoragemgmt.service	loaded	active	running	libstoragemgmt p
lvm2-monitor.service	loaded	active	exited	Monitoring of LV

lines 1-21

task2:

2. Change the default run level back to multi-user.target and reboot.

```
amiralsayed@amir:~$ systemctl set-default multi-user.target
Removed '/etc/systemd/system/default.target'.
Created symlink '/etc/systemd/system/default.target' -> '/usr/lib/systemd/system/multi-user.target'.
amiralsayed@amir:~$ S
```

task 3:

3. Send mail to the root user.

```
amiralsayed@amir:~$ mail root
Subject: task number 3 - lab 1 red hat system admin 2
To: root
this is a mail represting the task number 3 in todayes lab
it is a mail sent to root user
:__:
^D
-----
(Preliminary) Envelope contains:
To: root
Subject: task number 3 - lab 1 red hat system admin 2
Send this message [yes/no, empty: recompose]? yes
amiralsayed@amir:~$
```

task4:

4. Verify that you have received this mail.

```
amiralsayed@amir:~$ sudo mail -u root
[sudo] password for amiralsayed:
s-nail version v14.9.24. Type '?' for help
/var/spool/mail/root: 2 messages
> 1 amiralsayed      2025-12-28 13:58   16/565   "test mail to root"
> 2 amiralsayed      2025-12-28 14:12   18/637   "task number 3 - lab 1 red hat system admin 2"
& 2
[-- Message 2 -- 18 lines, 637 bytes --]:
Date: Sun, 28 Dec 2025 14:12:27 +0200
To: root@amir.localdomain
Subject: task number 3 - lab 1 red hat system admin 2
Message-Id: <20251228121227.33978C14BC82@amir.localdomain>
From: amiralsayed <amiralsayed@amir.localdomain>

this is a mail repressing the task number 3 in todayes lab
it is a mail sent to root user
:__:

&
At EOF
& 
```

task5:

5. Use systemctl utility to stop postfix service

```
amiralsayed@amir:~$ systemctl stop postfix.service
amiralsayed@amir:~$ systemctl status postfix.service
○ postfix.service - Postfix Mail Transport Agent
   Loaded: loaded (/usr/lib/systemd/system/postfix.service; disabled; preset: disabled)
   Active: inactive (dead)

Dec 28 14:12:27 amir postfix/cleanup[14313]: 33978C14BC82: message-id=<20251228121227.33978C14BC82@amir.locald>
Dec 28 14:12:27 amir postfix/qmgr[12325]: 33978C14BC82: from=<amiralsayed@amir.localdomain>, size=477, nrcpt=1>
Dec 28 14:12:27 amir postfix/local[14316]: 33978C14BC82: to=<root@amir.localdomain>, orig_to=<root>, relay=loc>
Dec 28 14:12:27 amir postfix/qmgr[12325]: 33978C14BC82: removed
Dec 28 14:26:16 amir systemd[1]: Stopping postfix.service - Postfix Mail Transport Agent...
Dec 28 14:26:16 amir postfix[15032]: postfix/postlog: stopping the Postfix mail system
Dec 28 14:26:16 amir postfix/postfix-script[15032]: stopping the Postfix mail system
Dec 28 14:26:16 amir postfix/master[12323]: terminating on signal 15
Dec 28 14:26:16 amir systemd[1]: postfix.service: Deactivated successfully.
Dec 28 14:26:16 amir systemd[1]: Stopped postfix.service - Postfix Mail Transport Agent.
lines 1-14/14 (END)
```

task 6:

6. Send mail again to the root user.

```
amiralsayed@amir:~$ mail root
Subject: trying to send main to root while postfix is down
To: root
i am trying to send a mail to root user while the mailing servese is down
waht i expect is that it won't be sent until the service is up agin, workign
like a queue.
^D
-----
(Preliminary) Envelope contains:
To: root
Subject: trying to send main to root while postfix is down
Send this message [yes/no, empty: recompose]? yes
amiralsayed@amir:~$
```

task 7:

7. Verify that you have received this mail.

```
amiralsayed@amir:~$ sudo mail -u root
s-nail version v14.9.24. Type '?' for help
/var/spool/mail/root: 2 messages
▶ 1 amiralsayed      2025-12-28 13:58  16/565  "test mail to root"
  2 amiralsayed      2025-12-28 14:12  18/637  "task number 3 - lab 1 red hat system admin 2"
&
```

task 8:

8. Use systemctl utility to start postfix service

```
amiralsayed@amir:~$ systemctl start postfix.service
amiralsayed@amir:~$ systemctl status postfix.service
● postfix.service - Postfix Mail Transport Agent
   Loaded: loaded (/usr/lib/systemd/system/postfix.service; disabled; preset: disabled)
   Active: active (running) since Sun 2025-12-28 14:28:33 EET; 6s ago
 Invocation: 4456f646f1884dfd8ab7b878e9684f1c
   Process: 15247 ExecStartPre=/usr/sbin/restorecon -R /var/spool/postfix/pid (code=exited, status=0/SUCCESS)
   Process: 15255 ExecStartPre=/usr/libexec/postfix/aliasesdb (code=exited, status=0/SUCCESS)
   Process: 15264 ExecStartPre=/usr/libexec/postfix/chroot-update (code=exited, status=0/SUCCESS)
   Process: 15265 ExecStart=/usr/sbin/postfix start (code=exited, status=0/SUCCESS)
 Main PID: 15333 (master)
    Tasks: 6 (limit: 98856)
   Memory: 4.1M (peak: 4.6M)
      CPU: 315ms
   CGroup: /system.slice/postfix.service
           └─15333 /usr/libexec/postfix/master -w
             └─15335 pickup -l -t unix -u
               └─15336 qmgr -l -t unix -u
                 └─15338 cleanup -z -t unix -u
                   └─15342 trivial-rewrite -n rewrite -t unix -u
                     └─15350 local -t unix

Dec 28 14:28:33 amir systemd[1]: Starting postfix.service - Postfix Mail Transport Agent...
Dec 28 14:28:33 amir postfix[15331]: postfix/postlog: starting the Postfix mail system
Dec 28 14:28:33 amir postfix/postfix-script[15331]: starting the Postfix mail system
Dec 28 14:28:33 amir postfix/master[15333]: daemon started -- version 3.8.5, configuration /etc/postfix
Dec 28 14:28:33 amir systemd[1]: Started postfix.service - Postfix Mail Transport Agent.
```

task 9:

9. Verify that you have received this mail.

```
amiralsayed@amir:~$ sudo mail -u root
s-nail version v14.9.24. Type '?' for help
/var/spool/mail/root: 3 messages 1 new
 1 amiralsayed      2025-12-28 13:58   16/565   "test mail to root"
 2 amiralsayed      2025-12-28 14:12   18/637   "task number 3 - lab 1 red hat system admin 2"
▶N 3 amiralsayed    2025-12-28 14:27   17/700   "trying to send main to root while postfix is down"
& 3
[-- Message 3 -- 17 lines, 700 bytes --]:
Date: Sun, 28 Dec 2025 14:27:39 +0200
To: root@amir.localdomain
Subject: trying to send main to root while postfix is down
Message-Id: <20251228122833.D01A3C14BC92@amir.localdomain>
From: amiralsayed <amiralsayed@amir.localdomain>

i am trying to send a mail to root user while the mailing servese is down
waht i expect is that it won't be sent until the service is up agin, workign
like a queue.

& []
```

10. Edit in the GRUB2 configuration file and change the timeout variable equal 20 seconds.
11. Edit in the GRUB2 configuration file and change your default operating system

[illegible]

```
amiralsayed@amir:~$ sudo grub2-mkconfig -o /boot/grub2/grub.cfg
[sudo] password for amiralsayed:
Generating grub configuration file ...
Found Windows Boot Manager on /dev/sdb2@efi/Microsoft/Boot/bootmgfw.efi
Adding boot menu entry for UEFI Firmware Settings ...
done
amiralsayed@amir:~$
```

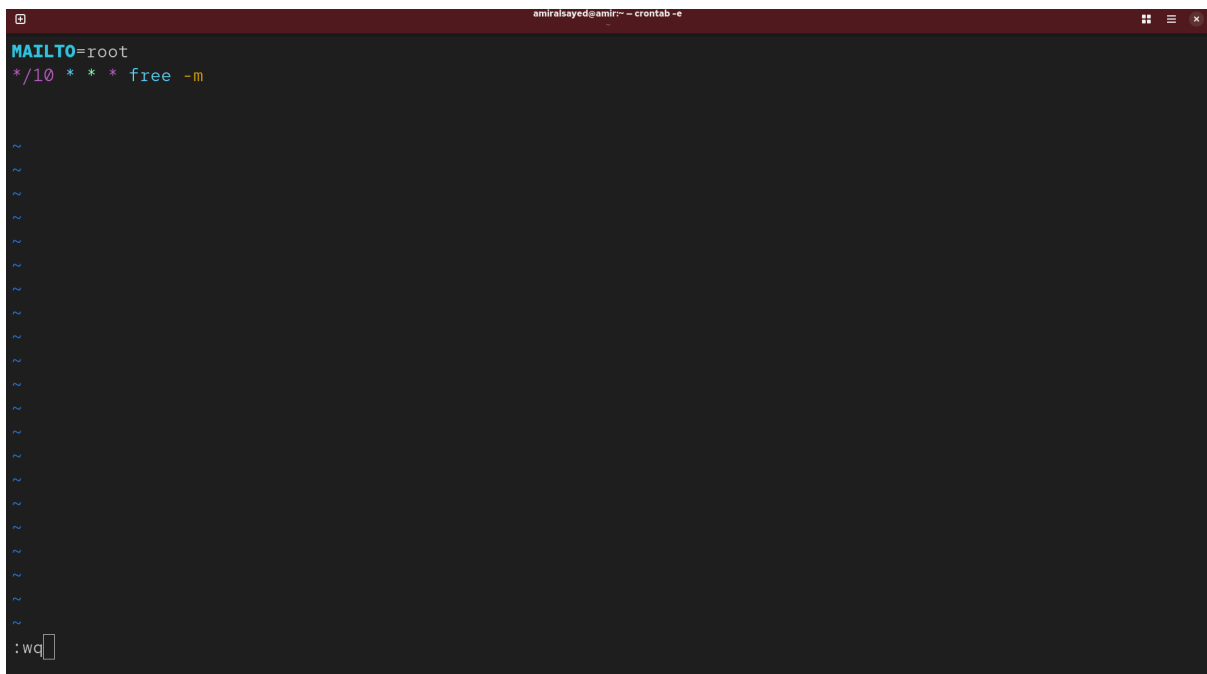
task 12, 13, 14,15:

12. You want to know some information about the status of the system every ten minutes today between the hours of 8:00 AM and 5:00 PM. to help investigate some performance issues you have been having. You suspect it might be memory related and want to keep an eye on those resources.

13. Use mail as the root user to check for e-mail from the cron jobs you have scheduled.

14. How could you send the output from these cron jobs to another e-mail address (the manager user)?

15. Use mail as the manager user to check for e-mail from the cron jobs you have scheduled.



The image shows a terminal window with a dark background. The title bar at the top reads "amiralsayed@amir:~ - crontab -e". The terminal content shows the configuration of a cron job. The first line is "MAILTO=root" in blue. The second line is a cron job entry: "* /10 * * * free -m", where the asterisks are in pink and "free -m" is in yellow. Below this, there are several tilde (~) characters representing the rest of the crontab file. At the bottom left, the prompt ":wq" is visible, indicating the user is in vi editor mode and ready to save and quit.

```
amiralsayed@amir:~ - crontab -e
MAILTO=root
*/10 * * * free -m
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
~
:wq
```

```
amiralsayed@amir:~$ sudo mail -u root
s-nail version v14.9.24.  Type '?' for help
/var/spool/mail/root: 6 messages 1 unread
 1 amiralsayed      2025-12-28 13:58  16/565  "test mail to root"
 2 amiralsayed      2025-12-28 14:12  18/637  "task number 3 - lab 1 red hat system admin 2"
 3 amiralsayed      2025-12-28 14:27  18/711  "trying to send main to root while postfix is down"
 4 amiralsayed      2025-12-28 15:14  15/488
 5 amiralsayed      2025-12-28 15:15  17/694
>U 6 amiralsayed      2025-12-28 15:44  17/693
& 6
[-- Message 6 -- 17 lines, 693 bytes --]:
Date: Sun, 28 Dec 2025 15:44:59 +0200
To: root@amir.localdomain
Message-Id: <20251228134500.05C1BC14BC82@amir.localdomain>
From: amiralsayed <amiralsayed@amir.localdomain>

total      used      free      shared buff/cache available
Mem:      15606      6111      4360       1008      6471      9494
Swap:      7995         0      7995

& 
```


task 16:

16. Write a script to print number of arguments, if the script was executed without any argument, log this instant to /var/log/scriptlogs

```
#!/bin/bash
if [ $# -eq 0 ]; then
    # Log to /var/log/scriptlogs if no arguments are provided
    echo "$(date): Script executed without arguments" >> /var/log/scriptlogs
else
    # Print the number of arguments
    echo "Number of arguments: $#"
```

"check_args.sh" 8L, 252B 8,2 All

```
amiralsayed@amir:~$ ./check_args.sh
amiralsayed@amir:~$ ./check_args.sh amir alsayed
Number of arguments: 2
amiralsayed@amir:~$ ./check_args.sh
amiralsayed@amir:~$ sudo cat /var/l
lib/ local/ lock/ log/
amiralsayed@amir:~$ sudo cat /var/log/scriptlogs
[sudo] password for amiralsayed:
Sun Dec 28 04:02:09 PM EET 2025: Script executed without arguments
Sun Dec 28 04:02:17 PM EET 2025: Script executed without arguments
amiralsayed@amir:~$
```